

Positive Recurrence for Bimolecular Weakly Reversible Stochastic Reaction Networks with Three Species and Two Linkage Classes

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Abstract

We prove positive recurrence on every closed communicating class for every finite weakly reversible stochastic mass-action network with molecularity at most two, at most three dynamically active species, at most two active linkage classes, and arbitrary positive rate constants. The proof combines an embedded labelled reaction-count Green construction with a finite source-rate compactification. The new stochastic ingredient is an aggregate physical workload-debt excursion: positive reactions are retained at their natural probabilities, their actual targets initiate faster return relaxations, and repeated slower arrivals are controlled by a scalar reflected debt process rather than by genealogical credits. A finite phase classification supplies service, structural exit, or an affine invariant. The one-active case is reduced to an exact polynomial-rate phase chain, while the two-active boundary geometry is closed by an independently reproduced finite atlas. Embedded return, physical return, and nonexplosion are treated separately.

Keywords. stochastic reaction network; weak reversibility; positive recurrence; mass-action kinetics; Green occupation; Foster–Lyapunov method; finite phase.

1 Model and theorem

Let the species set have at most three dynamically active coordinates. A complex is $y \in \mathbb{N}_0^d$, $d \leq 3$, and the network is *bimolecular* when $|y|_1 \leq 2$ for every complex. A labelled reaction channel is $e : y_e \rightarrow y'_e$, with rate $\kappa_e > 0$, reaction vector $\zeta_e = y'_e - y_e$, and stochastic mass-action propensity

$$\lambda_e(x) = \kappa_e(x)_{y_e}, \quad (x)_y = \prod_i \frac{x_i!}{(x_i - y_i)!}$$

when $x \geq y$, and zero otherwise. Parallel channels are retained. Each linkage class is a connected component of the undirected complex graph; weak reversibility means that each linkage class is strongly connected.

Fix one closed irreducible population class Γ . The population CTMC has formal generator

$$\mathcal{L}f(x) = \sum_e \kappa_e(x)_{y_e} [f(x + \zeta_e) - f(x)].$$

Theorem 1.1 (T3–2 positive recurrence). *Suppose the network has at most three dynamically active species and at most two active linkage classes. Then, for every positive rate vector, every closed*

communicating class is positive recurrent. Equivalently, every closed irreducible class has a unique stationary probability distribution.

The one-linkage case is supplied by the marked-target theorem [3]; earlier binary one-linkage results under additional structural assumptions appear in [2]. The present proof uses that theorem only at explicit one-linkage reductions.

2 Labelled embedded Green occupation

Let Z_n be the population immediately after the n -th reaction and E_{n+1} the actual labelled channel producing Z_{n+1} . Conditional on $Z_n = x$,

$$\mathbb{P}(E_{n+1} = e \mid Z_n = x) = \frac{\kappa_e(x)_{y_e}}{\Lambda(x)}, \quad \Lambda(x) = \sum_f \kappa_f(x)_{y_f}.$$

Fix $o \in \Gamma$, an increasing finite exhaustion $D_M \ni o$, and

$$T_o^+ = \inf\{n \geq 1 : Z_n = o\}, \quad \sigma_M = \inf\{n \geq 0 : Z_n \notin D_M\}, \quad \tau_M = T_o^+ \wedge \sigma_M.$$

Theorem 2.1 (Embedded labelled Green occupation). *If $\mathbb{E}_o T_o^+ = \infty$, then $t_M = \mathbb{E}_o \tau_M \rightarrow \infty$ and*

$$\nu_M(x, e) = \frac{1}{t_M} \mathbb{E}_o \sum_{n < \tau_M} \mathbf{1}\{Z_n = x, E_{n+1} = e\}$$

is a probability measure on labelled transitions. It escapes every fixed finite set. For bounded f ,

$$\sum_{x,e} \nu_M(x, e) [f(x + \zeta_e) - f(x)] = \frac{\mathbb{E}_o[f(Z_{\tau_M}) - f(o)]}{t_M}.$$

For every linear workload h ,

$$\sum_{x,e} \nu_M(x, e) h \cdot \zeta_e = h \cdot q_M, \quad q_M = \frac{\mathbb{E}_o[Z_{\tau_M} - o]}{t_M}.$$

The vectors q_M are bounded and every limit belongs to $\mathbb{R}_{\geq 0}^d$.

Proof. Monotone convergence gives $t_M \uparrow \mathbb{E}_o T_o^+ = \infty$. For a fixed state x , choose a finite positive-probability embedded path from x to o not revisiting x . Each visit to x has a fixed positive chance of terminating the killed excursion before the next visit, so its expected visit count is uniformly bounded in M . Division by t_M proves escape from finite sets.

The identities are pathwise telescoping. Since each reaction changes each coordinate by at most two, $\|Z_{\tau_M} - o\|_\infty \leq 2\tau_M$, so q_M is bounded. Also $q_{M,i} \geq -o_i/t_M$, proving nonnegativity of every limit. $\square$

3 Terminal chart localization

A Green occupation is a measure, not a representative sequence. We therefore compactify the occupation itself.

For each transition source record:

- (i) each population coordinate in $\mathbb{N}_0 \cup \{\infty\}$;
- (ii) capped availability $\min(x_i, 2)$ and enabled-source support;
- (iii) the actual current target and labelled channel;
- (iv) the complete recursively normalized physical source-propensity flag;
- (v) the tied positive ratio vector, workload chamber, and lattice phase.

The flag compactification is finite-depth: normalize the finite source vector; then normalize recursively on its zero coordinates. Thus every sequence has a convergent flag subsequence, including arbitrarily slow divergent species.

Lemma 3.1 (Terminal chart). *After subsequence extraction, every escaping Green occupation has a nonzero restriction to a terminal chart with fixed active set, fixed finite box for inactive coordinates, fixed enabled support, fixed source flag, compact tied ratio cell, fixed workload chamber and zero normalized structural-exit flux. For the positive integer workload $H(x) = h \cdot x$ of that chart,*

$$\sum_{x,e} \nu(x,e) h \cdot \zeta_e \geq 0. \quad (3.1)$$

Proof. Push ν_M to the compact descriptor. A weak limit exists and has total mass one at infinity. Bounded jumps preserve each asymptotic descriptor on a positive-mass limiting edge. Splitting over the finitely many active sets prevents, for example, mass near $(n, 0, 0)$ and $(0, n, 0)$ from being put in one chart that enables both active families.

Inactive coordinates have tight probability marginals. Choose a finite box carrying all but ε of the source and endpoint mass and pad by the maximal jump. Box crossings, support changes and flag refinements are retained as chart edges. Indicator telescoping gives a finite flow. Select a terminal strongly connected component of its support and diagonalize as $\varepsilon \downarrow 0$.

For (3.1), restrict to a long integer workload band. Summing lower-cut flux over consecutive cuts counts each bounded jump only a bounded number of times. Hence one lower cut has vanishing normalized inward flux. Upper-cut flux is outward. The finite chart-error tends to zero with ε . $\square$

A raw reaction-count normalization can be dominated by faster neutral motion. We therefore take exact physical traces at successively slower source layers. If a layer has bounded expected count, its bounded displacement cannot support escape. If its count diverges, normalize by that count and contract the already processed finite neutral phase. The source flag is finite, so this procedure terminates at the first occupied changing trace.

3.1 The iterated source simplex

For completeness we spell out the finite compactification used in Lemma 3.1. Let $\mathcal{S}_0$ be the finite set of source complexes enabled in the selected capped phase, and put

$$a_s(x) = \bar{\kappa}_s(x)_s, \quad s \in \mathcal{S}_0.$$

The first projective vector is

$$\rho_s^{(1)}(x) = \frac{a_s(x)}{\sum_{u \in \mathcal{S}_0} a_u(x)}.$$

At a boundary limit let $\mathcal{F}_1$ be its positive support. On $\mathcal{S}_1 = \mathcal{S}_0 \setminus \mathcal{F}_1$, provided it is nonempty, renormalize again:

$$\rho_s^{(2)}(x) = \frac{a_s(x)}{\sum_{u \in \mathcal{S}_1} a_u(x)}.$$

Continue. At most $|\mathcal{S}_0|$ steps are possible. The resulting ordered partition

$$\mathcal{F}_1 \gg \mathcal{F}_2 \gg \cdots \gg \mathcal{F}_r$$

and the positive ratio vector in every $\mathcal{F}_j$ constitute the complete flag. This is a closed subset of a finite product of simplices and is compact. Sources with vanishing first-order ratio are not discarded; they occur in a later $\mathcal{F}_j$.

The mass-action source exponents belong to the finite set

$$\{0, e_i, 2e_i, e_i + e_j\}.$$

After passing to one flag cell, its strict comparisons can be represented by a rational workload. Indeed, take the finitely many rational separating covectors produced by the successive normalized-log levels, clear denominators, and combine them with a sufficiently large integer base. The resulting $h \in \mathbb{Z}_{\geq 0}^d$ preserves every strict source comparison and is positive on every active coordinate. Equality of source orders is retained because the complete flag has already separated every nonzero lower-order difference.

3.2 Why chart flow is retained

Let $\mathcal{P}$ be a finite partition of one compact flag cell into its finite discrete data. For a part $A \in \mathcal{P}$, apply the bounded-test identity of Theorem 2.1 to $\mathbf{1}_A$. The normalized inflow minus outflow tends to zero. Hence every weak limit induces a finite circulation after source and return errors vanish. If a transition changes the active set, bounded box, source support, flag face or workload chamber, we place it on a directed chart edge rather than deleting it. A terminal strongly connected component therefore has zero limiting flux through every such edge.

The workload-band argument is also purely reaction-count based. If $J = \max_e |h \cdot \zeta_e|$, a transition crosses at most J integer lower cuts. Summing the normalized inward flux over K consecutive cuts is at most J . One cut has inward flux at most J/K . Letting $K \rightarrow \infty$ proves the local nonnegative outward balance without integrating an unbounded propensity.

4 Actual targets and aggregate workload debt

At an embedded jump, retain the target T_n of the actual labelled reaction. It is physically present. The inherited residual factorial identity is

$$V(x, t) = \sum_i \log((x_i - t_i)!), \quad \Delta V = \log \frac{(x)_t}{(x)_s}.$$

It is used only for same-linkage target paths beginning from an already carried target; no future activation is conditioned upon.

For the source-layer argument, scale the chart workload so that $H(y)$ is integer-valued on complexes. A positive reaction $y \rightarrow u$ has $H(u) > H(y)$ and hence creates an actual target in a strictly faster source layer.

Lemma 4.1 (Actual-target return prefix). *For every positive reaction $y \rightarrow u$, a directed path in its linkage from u back to y has a first target v with $H(v) \leq H(y)$. Every earlier source is strictly faster than y , and*

$$H(u - y) + H(v - u) = H(v - y) \leq 0. \quad (4.1)$$

The lifted path is sequentially enabled until a declared chart exit.

Proof. Strong connectivity supplies the path. The first-crossing definition gives the priority statement and (4.1). Each designated reaction produces the source of the next reaction, so the lifted path is physical. $\square$

The bounded coordinates, actual target, availability, source support, linkage and lattice data form a finite phase. Include every processed physical reaction and every return prefix.

Lemma 4.2 (Finite physical service phase). *A closed finite phase component without a negative workload edge contains no positive workload edge. Consequently each component is strict, edgewise zero, or has a structural exit.*

Proof. A positive edge has the return certificate of Lemma 4.1. Closedness retains that certificate. A finite path whose initial reward is positive and whose cumulative reward is nonpositive contains a negative edge. $\square$

Finite graph reachability and compactness of the tied-rate cell give constants $L < \infty$, $q > 0$, and $K < \infty$: in a strict component a negative edge is reached within L carrier changes with probability at least q ; the number of changes has geometric moments. Reactions leaving the selected source enabled do not delay its physical exponential clock. If the total unprocessed slower hazard is at most ε_R times the carrier hazard, then

$$\mathbb{P}\{\text{slower interruption before service}\} \leq K\varepsilon_R, \quad \varepsilon_R \rightarrow 0. \quad (4.2)$$

Fix an excursion baseline $B = H(X_0)$ and define only the physical scalar

$$D = (H(X) - B)^+.$$

Theorem 4.3 (Aggregate-debt clearing). *At successive service trials suppose*

$$D_{k+1} \leq (D_k - S_k)^+ + A_k,$$

and, on $D_k > 0$,

$$\mathbb{P}(S_k \geq 1 \mid \mathcal{F}_k) \geq p, \quad \mathbb{E}(A_k \mid \mathcal{F}_k) \leq a < p.$$

Then

$$\mathbb{E}\tau_0 \leq \frac{D_0}{p - a}.$$

If trial durations have conditional mean at most T , the physical clearing time has mean at most $TD_0/(p - a)$.

Proof. For integer $d > 0$,

$$(d - s)^+ + a - d \leq -\mathbf{1}_{\{s \geq 1\}} + a.$$

The conditional drift is at most $-(p - a)$. Sum through $n \wedge \tau_0$, use nonnegativity, and apply monotone convergence. $\square$

Here A_k is the workload of the first unprocessed slower event. Equation (4.2) gives $\mathbb{E}A_k = O(\varepsilon_R)$; choose the chart threshold so that it is below $p/2$. Multiple slower arrivals occur over repeated trials and are all represented in D .

After one service, if $D > 0$, restart from the new physical state and actual target. A strict phase gives the same service minorization; loss of support is an exit; a zero phase passes the unchanged debt to the next slower physical trace. No creator identity is retained.

At $D = 0$, a strict negative transition before the next positive event lowers the workload below the baseline. A failed trial creates bounded debt, which is cleared before the next zero-debt trial. Geometric trials therefore give genuine descent in finite mean.

4.1 The physical carrier graph

We give the construction behind the finite phase assertion. A carrier state records the actual target complex, the exact inactive-coordinate population, the capped active availability, the creator linkage and a position in one of the fixed return prefixes. It does not record an active population count. For each processed physical reaction there are two cases.

If the selected return source remains enabled, its exponential clock continues. Reactions not disabling that source are skipped in the carrier clock. If a reaction disables the source, then, inside a padded chart, it must consume one of the finitely many inactive cofactors required by the source. Its actual target and the new inactive population determine another carrier state. Loss of the padded box or active threshold is an exit.

Thus the carrier state space is finite. From every positive edge, the designated return prefix gives a path to cancellation or exit. Add all possible disabling transfers. A closed class avoiding cancellation would contain the positive edge but, by the return certificate, no negative edge, contradicting Lemma 4.2. This proves the qualitative reactivation alternative directly from physical channels.

For uniformity, eliminate source priorities recursively along the same complete flag. A strictly faster recurrent class either already contains service or is edgewise workload-neutral and is contracted by its finite Green/Poisson corrector. Consequently, when a carrier source at a given priority is inspected, only sources tied with it remain as genuine competitors; faster activity has already been absorbed into the finite phase.

Choose in every strict carrier class one path to service. There are finitely many phases and finitely many selected edges. Conditional channel probabilities from a fixed source depend only on its positive rate constants, and tied-source ratios range over a compact cell. Their minimum is a number $q > 0$. If the longest selected path has length L , one block succeeds with probability at least q^L . Repeated blocks give a geometric bound on carrier changes. This is a trace argument, not an assumption of fast mixing.

4.2 Slower arrivals

Let K bound the expected number of carrier changes in one strict trial. At every carrier phase, let λ_f be the total processed carrier hazard and λ_s the total unprocessed source hazard. The complete flag gives

$$\lambda_s \leq \varepsilon_R \lambda_f, \quad \varepsilon_R \downarrow 0.$$

Stop at the first unprocessed event. Compensator comparison over the carrier occupation gives

$$\mathbb{P}\{\text{interruption}\} \leq K\varepsilon_R. \quad (4.3)$$

If Δ_+ is the largest positive integer workload jump, then

$$\mathbb{E}A \leq K\Delta_+\varepsilon_R, \quad \mathbb{E}z^A \leq 1 + K\varepsilon_R(z^{\Delta_+} - 1), \quad z > 1. \quad (4.4)$$

There is at most one unprocessed event in a trial; multiple arrivals are handled over successive trials. This is why the arrival bound is independent of the current debt.

For a finite workload band, stop also at its padded upper boundary. One reaction overshoots that boundary by at most Δ_+ . A partial carrier excursion at the Green stopping time is therefore either a recorded upper-shell exit or has a uniformly bounded workload correction. When the layer trace count diverges, this one terminal correction vanishes after normalization. Thus no unfinished busy period is silently omitted.

4.3 Reactivation and zero debt

The service selector is recomputed after every trial from the current physical phase. In a strict component the same finite minorization applies. A zero component is carried to the next source layer, and any loss of the phase support is an exit. Therefore Theorem 4.3 can be restarted whenever $D > 0$, without identifying which reaction created the remaining excess.

At $D = 0$, run the same strict trial. A strict edge before the first unprocessed positive event gives a unit of descent below the baseline. On a failed trial, the first unprocessed event creates at most Δ_+ debt. Clear it by Theorem 4.3, then retry. The number of trials is geometrically dominated and all physical durations have finite mean.

5 Finite source-layer hierarchy

Process the complete source flag from fastest to slowest.

At the fastest layer, a positive edge is impossible: its target would be a faster source. Thus every recurrent class is strict or zero.

Inductively assume faster layers have been replaced by finite-mean effective transitions with nonpositive complete workload. Retain a raw event from the next layer at its natural probability. If it is positive, its actual target starts a faster debt excursion. Same-layer and slower interruptions enter the arrival variable in Theorem 4.3. Completion gives a nonpositive effective transition or a structural exit.

Proposition 5.1 (Source-layer alternative). *Every recurrent effective source-layer class has exactly one of:*

- (i) *a strict negative stationary corrected reward;*
- (ii) *edgewise zero reward and a bounded rational phase coboundary;*
- (iii) *positive structural-exit flux.*

The hierarchy terminates after finitely many layers.

Proof. All effective rewards are nonpositive. In a finite irreducible class, one negative edge has positive stationary flow and makes the mean strict. If the mean is zero, every positive-probability edge has reward zero. Cycle sums vanish, so path sums define the phase coboundary. A layer of vanishing fraction is retained as the next slower trace. There are finitely many source layers. $\square$

If every layer is zero, $H(X) + \psi(\phi)$ is invariant on the terminal recurrent support. The phase term is bounded and h is positive on every active coordinate, contradicting escape. Because zero is edgewise after complete faster relaxation, the corrected variance is also zero; no hidden critical random walk remains.

5.1 Induction inside one raw source layer

The induction has an inner priority induction. At the fastest source priority, a positive reaction is impossible because its target would be a strictly faster source. Hence the fastest raw edges are neutral or negative.

Suppose priorities above p have already been replaced by nonpositive finite-mean effective transitions. A positive raw reaction with source priority p produces an actual target of priority greater than p . Start the carrier process immediately after that actual reaction, while retaining the reaction's positive workload in D . A second event from priority p or below is an unprocessed arrival. Equations (4.3)–(4.4) and Theorem 4.3 show that the entire positive event plus faster relaxation has nonpositive endpoint workload in finite mean, or exits the chart. It may cancel exactly; it is not called strict descent.

Descending through the finite priority set converts every raw edge of the source layer into a nonpositive effective edge. Only then is the finite recurrent-class alternative of Proposition 5.1 applied.

5.2 Passage to a slower reaction-count trace

Let $C_j(M)$ be the expected number of layer- j events in the killed Green path. If $C_j(M)$ is bounded, its bounded jumps have zero contribution to an escaping endpoint after division by the relevant divergent count. If $C_j(M) \rightarrow \infty$, normalize its exact labelled trace, including the complete faster excursions between successive layer- j events. Positive chart-exit counts are retained. This gives a nonzero finite-phase occupation at layer j .

Thus “zero at the current normalization” never means deletion. It means that the same physical events appear at the next slower normalization. Since the source set is finite, the procedure cannot continue indefinitely.

5.3 The all-zero branch

Suppose every effective layer is edgewise zero. On each finite recurrent phase class, define ψ by integrating $-h \cdot \Delta X$ along phase paths. Zero cycle sums make this well-defined. Correctors from successive layers add, and only finitely many are present. Hence

$$h \cdot X + \psi(\phi)$$

is invariant for every transition of the terminal physical trace. Events outside the trace have bounded count or positive exit flux. The first cannot produce an escaping endpoint and the second contradicts terminality. Since ψ is bounded, the invariant excludes escape in an active direction.

6 Active-coordinate cases

6.1 Three active coordinates

Every possible bimolecular source is enabled. At the first occupied changing layer, every changing reaction is nonincreasing and at least one is strict: a positive reaction at the highest changing source level would force, along its linkage return cycle, a negative reaction sourced at a still higher level. Compact tied ratios give strictly negative physical flux, contrary to (3.1). If no layer changes H , it is a global invariant.

6.2 One active coordinate

Let $A = n$ be active and let B, C remain in a finite box. Every transition is

$$(n, \phi) \rightarrow (n + k, \phi'), \quad |k| \leq 2,$$

with rate

$$c_2(\phi, e)n(n-1) + c_1(\phi, e)n + c_0(\phi, e).$$

If $2A$ is a source, every genuine reaction from it lowers A , while upward reactions have degree at most one:

$$\mathcal{L}A \leq -c_2A(A-1) + c_1A + c_0, \quad c_2 > 0.$$

Without $2A$, every degree-one edge has increment zero or minus one. Finite recurrent degree-one classes have strict negative mean and a Poisson corrector, or are edgewise zero and are contracted. A degree-zero birth creates at most one A and an actual target A or $A + D$; the creator linkage provides its service path and the aggregate-debt theorem includes every bounded-coordinate interruption.

In an all-zero class, let K be the bounded service species occurring in one- A complexes. Either a unary one- A complex or a zero- A complex containing K gives unpaired service, or

$$A - \sum_{D \in K} D$$

is an exact invariant. Since the bounded coordinates and phase correction are bounded, escape is impossible.

6.2.1 Exact finite-phase correction

Let Q_1 be the finite generator formed by the coefficients of n in the degree-one phase transitions. For an irreducible recurrent class let π be its stationary probability and $g(\phi) \leq 0$ its conditional mean A -increment. If one negative edge is present, then $\bar{g} = \pi g < 0$. Solve

$$Q_1\chi = -(g - \bar{g})$$

with one normalization. Since the phase is finite, χ is bounded and

$$\mathcal{L}[A + \chi(\phi)] = n\bar{g} + O(1),$$

uniformly over the bounded-coordinate phase.

If $\bar{g} = 0$, nonpositivity forces every positive-flow degree-one edge to have zero increment. Contract the class. Degree-zero events and their complete fast relaxations form a finite macrochain. A positive birth is kept at its natural rate; its actual one- A target starts the faster carrier process. Repeated births are handled by aggregate debt.

In an edgewise-zero macroclass, the service-token dichotomy is exact. If there is no unary one- A complex and no zero- A complex contains a service species from a binary one- A complex, then every one- A complex carries exactly one such token and every zero- A complex carries none. Therefore $A - \sum_{D \in K} D$ is preserved reaction by reaction. Otherwise the finite phase contains an unpaired service or an exit. Thus a zero mean with nonzero variance is never classified as an invariant.

6.3 Two active coordinates and the finite atlas

We first record the interface between the stochastic trace and the finite classification. Call a linkage *available* when it falls in one of the available branches of the certified top-complex alternative. In the two-active chart, an available linkage has one of the following witnesses: a top source containing two active particles, a unary top source, or an active-plus- C target whose service cofactor is present in a lower complex. The first two witnesses are enabled throughout the padded chart. In the third case the exact finite C -phase either restores the cofactor and enables service, or leaves the box/support. Actual-target return prefixes and the carrier-race lemma therefore give strict service or exit. Consequently, a closed edgewise-zero terminal phase can contain only flat or shielded linkages. If both linkages represented in it are shielded, the pair atlas below applies.

Let A, B be active and C bounded. Up to exchanging A, B , four workloads cover every chamber and wall:

$$(1, 1, 0), \quad (2, 3, 0), \quad (1, 2, 0), \quad (1, 3, 0).$$

Three independent implementations exhaust all assignments of the ten complexes to linkage 1, linkage 2 or unused:

workload assignments	187,488
shielded assignments	446
common affine invariants	382
deficiency-zero assignments	60
ordered service assignments	4
unclassified	0.

Available linkages are handled by Proposition 5.1. A closed shielded phase is therefore one of:

- (i) a common affine invariant positive on A, B ;
- (ii) a weakly reversible deficiency-zero network;
- (iii) one of the two service architectures

$$\{C, 2C\} \ \& \ \{0, A, 2A, B + C\},$$

$$\{0, C, 2C\} \ \& \ \{A, 2A, B + C\}.$$

The deficiency-zero branch has a summable product-Poisson stationary measure. In the service branch $W = B - C$; the mixed linkage preserves W , while a finite-shell C -trial gives

$$W_\tau \leq W_0 - 1$$

or a declared promotion/exit.

7 Proof of the main theorem

Assume, for contradiction, infinite expected positive return count. Apply Theorem 2.1 and Lemma 3.1. The terminal chart has nonnegative outward workload balance.

If it has three active coordinates, the strict source-layer flux is negative. If it has two, Proposition 5.1 or the certified atlas gives negative flux, an invariant, product-form stationarity, or service descent. If it has one, the polynomial phase analysis gives quadratic descent, strict corrected linear descent, an invariant, or exit. With no active coordinates the chart is finite. Every possibility contradicts terminal escape. Therefore the embedded chain has finite expected positive return count.

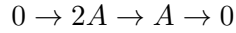
At every nonabsorbing population state, some enabled falling factorial is a positive integer. Hence $\Lambda(x) \geq \kappa_{\min} > 0$, and finite expected embedded return count implies finite expected physical return time.

Population-increasing reactions have source molecularity at most one, so their aggregate rate is bounded by $C(1 + |X|_1)$. A linear pure-birth comparison bounds upward jumps on finite time intervals. Between upward jumps total population is nonincreasing and the chain remains in a finite shell, where total rate is bounded. Thus the CTMC is nonexplosive.

The standard irreducible countable-state equivalence now gives positive recurrence and uniqueness of the stationary probability [4]. This proves Theorem 1.1. $\square$

A Forbidden activation regressions

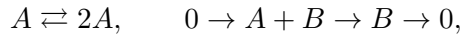
The proof never conditions on a future activation. For



the conditioned activation/path block satisfies

$$J_n = \frac{7 \log n + 1}{n^2} + O\left(\frac{\log n}{n^3}\right) > 0.$$

For



conditioning on $0 \rightarrow A + B$ and appending only the target episode is asymptotic to $+\log n$. Both invalid decompositions are preserved as regression tests. In the proof, every activation remains in its preceding unconditioned physical source-layer trace.

B Verification boundary

Computation is not used to prove the infinite-state Green or debt theorems. It independently checks the finite atlas, priority graph alternatives, one-active source signs, exact queue inequalities, Poisson arithmetic, service supports, and the two activation counterexamples.

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